

M13.8 Pitot-Static Systems 408-426 — Full Set

81 questions · Level 1: 37 · Level 2: 38 · Level 3: 6

LTT Module 13.8, pages 408-426. Grouped by level. Three answers each, one correct, in the Part-66 exam format.
No calculations, no unit conversions, no worked examples from the manual, nothing aircraft-type specific.

Pitot-static systems

Level 1 — recall

1. Which pressure is used by an altimeter
 - a. Pitot and static pressure
 - b. Pitot pressure only
 - c. Static pressure only
2. An airspeed indicator in a simple system is connected to two pressure lines. Which pressures does it use?
 - a. Pitot pressure and static pressure
 - b. Pitot pressure and total pressure
 - c. Static pressure and differential pressure
3. A small aircraft is fitted with one pitot system and one static system. Which three instruments are supplied by it?
 - a. Altimeter, airspeed indicator and total air temperature gauge
 - b. Altimeter, vertical speed indicator and standby compass
 - c. Altimeter, vertical speed indicator and airspeed indicator
4. How does pitot pressure reach the relevant instruments in a simple system
 - a. It is sensed by a pitot probe and guided via a tube to the instruments
 - b. It is sensed by a static port and guided via a tube to the instruments
 - c. It is sensed by a pitot probe and guided via a wind vane to the instruments
5. Which instrument indicates the altitude of an aircraft
 - a. The altimeter
 - b. The airspeed indicator
 - c. The vertical speed indicator
6. Which pressure is used by a vertical speed indicator
 - a. Pitot and static pressure
 - b. Static pressure only
 - c. Pitot pressure only
7. An airspeed indicator is fitted in a small aeroplane. What does this instrument indicate?
 - a. The altitude of the aircraft
 - b. The speed in the surrounding air
 - c. The vertical speed of the aircraft
8. How many pitot systems and static systems does the simple system of a small aircraft have
 - a. One pitot system and two static systems
 - b. One pitot system and one static system
 - c. Two pitot systems and one static system
9. Where are the pressure measuring locations of one static pressure system situated
 - a. On the nose and the tail of the fuselage
 - b. On the upper and lower side of the fuselage
 - c. On the left and right hand side of the fuselage
10. How many separate pitot-static pressure systems are installed on modern commercial aircraft
 - a. One or two separate systems
 - b. Three or four separate systems
 - c. Two or three separate systems
11. A maintenance manual lists the separate pitot-static pressure systems of a large aeroplane. How are they termed according to their allocation?
 - a. Captain's system, F/O's system and auxiliary system
 - b. Captain's system, purser's system and auxiliary system
 - c. Captain's system, F/O's system and alternate system
12. In which small aircraft is the static source selector valve also found
 - a. Small single pilot aircraft certified for visual flight rules
 - b. Small two pilot aircraft certified for instrument flight rules
 - c. Small single pilot aircraft certified for instrument flight rules
13. What does IFR stand for in the description of the static source selector valve
 - a. Instrument flight rules
 - b. Inertial flight rules
 - c. Instrument flight reference

14. What is static pressure
 - a. The build-up of air pressure in a tube facing the airstream
 - b. The actual atmospheric pressure surrounding the aircraft
 - c. The difference between total pressure and ram pressure
15. Which authority introduced the international standard atmosphere
 - a. The international aviation authority ICAO
 - b. The international aviation authority EASA
 - c. The international aviation authority IATA
16. A pneumatic device is calibrated to the international standard atmosphere. Which value of air density at mean sea level applies?
 - a. 1,150 kg/m³
 - b. 1,013 kg/m³
 - c. 1,225 kg/m³
17. What are the ISA values of temperature and air pressure at mean sea level
 - a. 15 °C and 1225 mbar
 - b. 20 °C and 1013 mbar
 - c. 15 °C and 1013 mbar
18. Where on the aircraft are the static ports fitted, and how do they sit in the skin?
 - a. Protruding from the skin on the sides of the aircraft
 - b. Flush with the skin on the nose of the aircraft
 - c. Flush with the skin on the sides of the aircraft
19. What do some aircraft types have fitted at their static ports
 - a. Pneumatically heated static ports
 - b. Electrically drained static ports
 - c. Electrically heated static ports
20. Where does the static pressure originate on the aircraft
 - a. In the static ports
 - b. In the pitot probe
 - c. In the drain hole
21. At which level are the ISA values of density, temperature and pressure defined
 - a. At the cruise level
 - b. At mean sea level
 - c. At the tropopause
22. The current air pressure has to reach the individual air data measuring systems. How is it carried there?
 - a. Via hoses and pipes
 - b. Via a digital converter
 - c. Via a wind vane
23. Most helicopters carry an extra item at their static ports. What is this item?
 - a. Special drain holes
 - b. Special cover plates
 - c. Special flapper plates
24. What does the abbreviation SSE stand for
 - a. Standard source error
 - b. Static system error
 - c. Static source error
25. What is pitot pressure
 - a. The actual atmospheric pressure in the air surrounding the aircraft
 - b. The build-up of air pressure in a tube with its opening facing the airstream
 - c. The build-up of air pressure in a tube with its opening across the airstream
26. Which two pressures make up the total pressure P_{tot}
 - a. Ram pressure and differential pressure
 - b. Static pressure and differential pressure
 - c. Ram pressure and static pressure
27. Which quantity does the symbol q represent in the pitot pressure relation
 - a. Ram pressure
 - b. Total pressure
 - c. Static pressure

28. Where is the pitot tube usually located on the aircraft
- Near the nose, on the wing leading edge or below the flaps
 - Near the tail, on the fin itself or below the rudder
 - Near the nose, on the nose itself or below the cockpit
29. What does the pitot tube consist of and which pressure is present in it during the entire flight
- A cylindrical pipe in which total pressure P_{tot} is present
 - A cylindrical pipe in which static pressure P_s is present
 - A flapper plate chamber in which total pressure P_{tot} is present
30. What does electrical heating of the pitot tube prevent
- The tube from freezing up
 - The tube from draining
 - The tube from deforming
31. How is the pressure in the pitot tube transferred to the interior of the aircraft
- Via piping
 - Via a baffle
 - Via a drain hole
32. Why is the Prandtl tube electrically heated
- To prevent condensation
 - To prevent deformation
 - To prevent icing
33. A Prandtl tube supplies both pressures from one probe. Where is it located on the aircraft?
- At the rear of the aircraft
 - Under the wing of the aircraft
 - At the front of the aircraft
34. Through which part of the Prandtl tube does the total pressure enter
- The holes in the sides
 - The flapper plate
 - The cylindrical inlet
35. An air data system receives three parameters from one single multi function probe. Which parameters are they?
- Angle of attack, static air temperature and static pressure
 - Angle of sideslip, total air temperature and total pressure
 - Angle of attack, total air temperature and total pressure
36. Which part of the multi function probe measures the angle of attack
- A pitot probe
 - A TAT tube
 - A wind vane
37. Which part of the multi function probe measures the total air temperature
- The wind vane of the probe
 - The tube of the probe
 - The pitot probe element

Level 2 — understand and explain

38. Why are the left and right hand static pressure measuring locations of one static system connected to each other
- To drain any water collected inside the static pressure lines
 - To compensate for any unbalance in pressure measurement
 - To increase the pressure delivered to the connected instruments
39. Why are modern commercial aircraft fitted with separate pitot-static pressure systems
- For safety reasons
 - For calibration reasons
 - For weight reasons
40. What does the alternate static source selector valve allow the Captain to do
- Use the copilot's static pressure system
 - Use the copilot's pitot pressure system
 - Isolate the copilot's static pressure system

41. The Captain's static system becomes faulty in flight. Which statement about switching over to the auxiliary system by means of the static air source selector valve is correct?
- On every aircraft type the indicators can be switched to the auxiliary system
 - Only on a few aircraft types can the indicators be switched to the auxiliary system
 - Only on a few aircraft types can the indicators be switched to the pitot system
42. On which two quantities does air pressure p depend
- Air density and temperature
 - Air density and mach number
 - Temperature and angle of attack
43. What is done with the ISA values defined at mean sea level
- All pneumatic devices are corrected according to the flap position
 - All pneumatic devices are calibrated according to these values
 - All pneumatic devices are heated according to these values
44. What do aircraft manufacturers ensure when they choose the position of the static ports
- That they lie in areas with low temperature and air density variation
 - That they lie in areas with low turbulence and ram air distortion
 - That they lie in areas with high turbulence and ram air distortion
45. A technician is told that the static port zones must be absolutely clean and have a smooth surface. Which reason does the warning give?
- So that no electrical heating current can be lost
 - So that no pressure distorting turbulences can occur
 - So that no water can enter the connected pressure lines
46. Why do most helicopters have special cover plates at the static ports
- To compensate the effects caused by the strong downwash of the rotor
 - To compensate the effects caused by the strong vibration of the rotor
 - To compensate the effects caused by the high angle of attack of the rotor
47. Why was the international standard atmosphere introduced
- To have a uniform heating value for the probes
 - To have a uniform reference value for aviation
 - To have a correction table for high speed flight
48. Why can a static source error arise on an aircraft
- It is not always possible to find turbulence and ram air free zones on the aircraft
 - It is not always possible to keep the static ports electrically heated in flight
 - It is not always possible to calibrate the pneumatic devices to the ISA values
49. An engineer is told that the static source error of a jet aeroplane is not a constant value. On which factors does it depend?
- The mach number, the air density, the flap position and the outside air temperature
 - The mach number, the angle of attack, the flap position and the nose gear position
 - The mach number, the angle of attack, the rudder position and the nose gear position
50. How large is the static source error on slow-flying propeller planes
- So small that it is negligible
 - So small that it must be measured
 - So large that it must be corrected
51. The crew has to obtain the correct value from the distorted values being displayed. Where are the correction tables found?
- In the Aircraft Maintenance Manual or in the flight log
 - In the Operating Manual or in the flight regulations
 - In the Operating Manual or in the weight and balance data
52. What can make the static source error of one individual aircraft change over time
- Recalibration changes to the air data computer
 - Surface changes to the static ports
 - Surface changes to the pitot probe inlet
53. Which pressure reading is affected by a static source error
- The ram pressure reading
 - The total pressure reading
 - The static pressure reading

54. Which of these affects the static source error even though you would not expect it to?
- The position of the flaps
 - The position of the nose gear
 - The angle of attack
55. What can happen to the static source error of a jet aeroplane flying near the speed of sound
- It can exceed the tolerances required by law
 - It can fall below the tolerances required by law
 - It can become independent of the angle of attack
56. What do the correction tables enable the crew to do
- Calibrate the pneumatic devices to the ISA values
 - Find the correct value from the distorted values displayed
 - Switch the indicators over to the auxiliary system
57. Two aeroplanes of the same type show different static source errors. Which statement matches the manual?
- Deviations can occur only between different aircraft types
 - Deviations can occur within any one aircraft type
 - Deviations can occur only on slow-flying propeller planes
58. Which pressure alone is required to measure the speed of the aircraft in relation to the ambient air
- The ram pressure q
 - The total pressure P_{tot}
 - The static pressure p_s
59. How is ram pressure q obtained in the instrument
- By measuring the static pressure at two separate static ports
 - By measuring the sum of the total pressure and the static pressure
 - By measuring the differential pressure between total and static pressure
60. Ram pressure cannot be sensed on its own. Which pressure is already known from the static ports and is used to obtain it?
- The total pressure P_{tot}
 - The dynamic pressure q
 - The static pressure p_s
61. A pitot tube is being refitted to an aeroplane after removal. How must it be aligned?
- So that it faces the predetermined airstream
 - So that it faces the predetermined rotor downwash
 - So that it lies across the predetermined airstream
62. What do the small drain hole and the baffle of a pitot tube ensure
- That ram air entering the tube is slowed down
 - That water entering the tube is drained away
 - That the tube inlet is kept free of any deformation
63. A student is worried that the drain hole in a pitot tube lets pressure escape. Which statement from the manual answers this?
- The drain hole and the baffle do not affect the total pressure
 - The drain hole and the baffle slightly reduce the total pressure
 - The drain hole and the baffle slightly reduce the static pressure
64. During a walk-around a technician finds a small dent at the inlet of a pitot tube. Which instruction from the manual applies?
- The tube inlet may display slight deformation if the heating is serviceable
 - The tube inlet must not display any signs of deformation whatsoever
 - The tube inlet may display slight deformation if the drain hole is clear
65. What does the Prandtl tube combine in one unit
- A pitot tube with a sensor to measure total air temperature
 - A pitot tube with a sensor to measure static pressure
 - A static port with a sensor to measure the angle of attack
66. A Prandtl tube has a cylindrical inlet at the front and a row of holes drilled in its sides. What do these side holes supply?
- Information for one or two connected pitot lines
 - Drainage for one or two connected water traps
 - Information for one or two connected static lines
67. What do the flapper plate and drain hole of a Prandtl tube prevent
- Water from entering the pitot line
 - Icing from forming at the tube inlet
 - Roughness from forming on the tube surface

68. A Prandtl tube is found with a rough surface after a repair. Which consequence does the manual warn about?
- It will lead to heating element failures
 - It will lead to total pressure errors
 - It will lead to static pressure errors
69. A technician is about to touch a heated pitot or Prandtl tube shortly after the aircraft has landed. Which warning applies?
- Risk of shock
 - Risk of icing
 - Risk of burns
70. How does a Prandtl tube obtain the ram pressure and static pressure needed for speed indication
- From two separate measuring points
 - From one single ram air tube
 - From two connected static ports
71. Which two separate measuring points does the Prandtl tube replace
- The pitot tube and the static port
 - The static port and the difference bellows
 - The pitot tube and the air data computer
72. A Prandtl tube delivers total pressure and static pressure. Where is the required differential pressure obtained?
- In the tube itself or in the piping
 - In the static port or in the drain hole
 - In the instrument or in a computer
73. A multi function probe must deliver the total pressure to the air data system. Which of its parts provides this parameter?
- The pitot probe
 - The temperature tube
 - The wind vane
74. Each multi function probe has to pass its measured data to the related computer. Through which item does it do so?
- Through the air data computer
 - Through the digital converter
 - Through the difference bellows
75. To which item does a multi function probe finally send its measured data
- The related computer
 - The alternate static valve
 - The heated static port

Level 3 — apply and reason

76. A technician traces one static pressure system of a small aeroplane and finds a measuring location on the left and on the right hand side of the fuselage. Which statement describes this arrangement correctly?
- Each location belongs to a separate static system and they stay isolated
 - Both locations belong to the same pitot system and are connected together
 - Both locations belong to the same static system and are connected together
77. An aeroplane has a fault in one pilot's static system and is not one of the few types that can be switched to the auxiliary system. Which possibility remains?
- Use of the pitot system via the selector valve
 - Use of the auxiliary system via the selector valve
 - Use of the other pilot's static system via the selector valve
78. Why are static ports placed in areas with low turbulence and ram air distortion
- Because turbulence and ram air distortion block the static pressure hoses
 - Because turbulence and ram air distortion overheat the static port zones
 - Because turbulence and ram air distortion falsify the static pressure reading
79. A jet aeroplane flies near to the speed of sound and its static source error exceeds the tolerances required by law. Which correction becomes necessary?
- A correction using a differential pressure bellows
 - A correction using an air data computer
 - A correction using an alternate static source valve
80. A static source error falsifies the static pressure supplied to a simple pitot-static system. Which instruments are affected?
- All three instruments of the simple system
 - Only the vertical speed indicator of the system
 - Only the airspeed indicator of the system

- 81.** A slow-flying propeller aeroplane and a jet aeroplane near the speed of sound both have a static source error. Which comparison is correct?
- a.** Beyond the legal tolerance on both aeroplanes and corrected by the air data computer
 - b.** Negligible on the propeller aeroplane and possibly beyond the legal tolerance on the jet
 - c.** Negligible on the jet aeroplane and possibly beyond the legal tolerance on the propeller plane

Answer key

No.	Ans	Level	LTT page	Why
1	c	L1	58	The manual states the altimeter works on static pressure alone.
2	a	L1	58	The airspeed indicator is the only one of the three that needs both pitot and static pressure.
3	c	L1	58	The simple system feeds exactly these three pitot-static instruments.
4	a	L1	58	A pitot probe senses the pressure and a tube carries it to the instruments.
5	a	L1	58	The altimeter is the instrument the manual gives for altitude.
6	b	L1	58	The manual gives static pressure as the only supply of the vertical speed indicator.
7	b	L1	58	The manual defines the airspeed indicator as showing the speed in the surrounding air.
8	b	L1	58	The simple system has exactly one of each.
9	c	L1	58	Each static system has one measuring location on each side of the fuselage.
10	c	L1	60	The manual gives two or three separate systems.
11	a	L1	60	These are the three names given in the manual.
12	c	L1	60	The manual names single pilot aircraft certified for IFR.
13	a	L1	60	The manual spells IFR out as instrument flight rules.
14	b	L1	62	Static pressure is simply the ambient atmospheric pressure around the aircraft.
15	a	L1	62	ICAO introduced the ISA.
16	c	L1	62	The ISA sea level density given in the manual is 1,225 kg/m ³ .
17	c	L1	62	The manual defines 15 °C and 1013 mbar at MSL.
18	c	L1	62	Static ports sit flush in the skin on the sides of the fuselage.
19	c	L1	62	The manual states that some types use electrically heated static ports.
20	a	L1	62	The static ports are the source of static pressure.
21	b	L1	62	The manual gives the ISA figures for mean sea level.
22	a	L1	62	Hoses and pipes carry the sensed pressure to the measuring systems.
23	b	L1	62	The manual names special cover plates.
24	c	L1	64	SSE is the static source error.
25	b	L1	66	Pitot pressure builds up in a tube whose opening points into the airstream.
26	c	L1	66	The manual gives total pressure as ram pressure plus static pressure.
27	a	L1	66	The manual defines q as the ram pressure.
28	c	L1	68	These are the three locations the manual names.
29	a	L1	68	It is a cylindrical pipe carrying total pressure throughout the flight.
30	a	L1	68	The heater is there to stop the tube icing up.
31	a	L1	68	Piping carries the pressure from the tube into the aircraft.
32	c	L1	70	Electrical heating keeps the probe free of ice.
33	c	L1	70	It sits at the front, like the pitot tube.
34	c	L1	70	Total pressure enters at the cylindrical inlet at the front.
35	c	L1	72	AOA, TAT and total pressure are the three outputs of the probe.
36	c	L1	72	The wind vane is the AOA sensing element.
37	b	L1	72	A tube inside the probe senses the total air temperature.
38	b	L2	58	Cross-connecting the two sides evens out an unbalanced pressure reading between them.
39	a	L2	60	Redundant systems are installed for safety.
40	a	L2	60	The valve switches the Captain over to the copilot's static source.
41	b	L2	60	The manual restricts this switching possibility to a few aircraft types only.
42	a	L2	62	The manual lists air density and temperature as the two quantities.
43	b	L2	62	The ISA values are the calibration reference for pneumatic devices.
44	b	L2	62	Low turbulence and low ram air distortion keep the static reading true.
45	b	L2	62	Dirt or roughness creates turbulence that distorts the sensed pressure.
46	a	L2	62	The cover plates offset the pressure effects of the rotor downwash.
47	b	L2	62	The ISA gives aviation one common reference.
48	a	L2	64	Where no turbulence free and ram air free zone exists, the sensed static pressure is falsified.
49	b	L2	64	These four items are the dependencies the manual lists for the SSE.
50	a	L2	64	At low speed the error is negligible.

51	b	L2	64	The manual names the Operating Manual and the flight regulations as the sources of the correction tables.
52	b	L2	64	Changes to the static port surfaces alter the error over the life of the aircraft.
53	c	L2	64	The SSE falsifies the static pressure reading.
54	b	L2	64	Flaps and angle of attack are the obvious influences; the nose gear's position is the one the manual singles out as unusual.
55	a	L2	64	Close to the speed of sound the error may go beyond the legal tolerance.
56	b	L2	64	The tables turn a distorted indication back into the correct value.
57	b	L2	64	The manual allows for differences between individual aircraft of the same type.
58	a	L2	66	Only the dynamic or ram pressure q is needed for speed measurement.
59	c	L2	66	Ram pressure is the difference between total and static pressure.
60	c	L2	66	The static ports supply p_s , which is subtracted from P_{tot} to leave q .
61	a	L2	68	The tube must point into the airstream to build up total pressure.
62	b	L2	68	Their job is to get rid of water that has entered the tube.
63	a	L2	68	The manual states in brackets that they do not affect the total pressure.
64	b	L2	68	The manual allows no deformation of the inlet at all.
65	b	L2	70	It puts the pitot function and a static pressure sensor into one probe.
66	c	L2	70	The side holes feed the static lines, while the front inlet feeds total pressure.
67	a	L2	70	They keep water out of the pitot line.
68	c	L2	70	Roughness disturbs the flow over the side holes and falsifies static pressure.
69	c	L2	70	The heated probe stays hot, so the manual warns of burns.
70	b	L2	70	The Prandtl tube takes both pressures at one single tube.
71	a	L2	70	The manual names the pitot tube and the static port as the two points it replaces.
72	c	L2	70	The subtraction is done in the instrument or in a computer, not in the probe.
73	a	L2	72	The built-in pitot probe supplies the total pressure.
74	b	L2	72	The digital converter is the link between the probe and the related computer.
75	a	L2	72	The data ends up at the related computer.
76	c	L3	58	One static system uses both side locations and they are joined to each other.
77	c	L3	60	Switching to the auxiliary system is limited to a few types, so the cross-over to the other pilot's static system is what remains.
78	c	L3	62	Where such zones cannot be found the static pressure reading is affected, so a quiet zone keeps the reading true.
79	b	L3	64	The manual calls for an air data computer correction in this case.
80	a	L3	64	All three instruments of the simple system are supplied with static pressure, so all three are affected.
81	b	L3	64	The manual calls the error negligible at low speed and possibly beyond the legal tolerance near the speed of sound.